Mecánica teórica — clase # 7 — 19/08/2025

$$\vec{r}_a = \vec{r}_a(q_i, t) \; ; \quad i = 1, 2, 3N = n \qquad m_a \frac{d^2\vec{r}_a}{dt^2} = \vec{F}_a$$

$a = 1, 2, \ldots, N$

$$\sum_{a=1}^{N} m_a \frac{d^2\vec{r}_a}{dt^2} \cdot \delta\vec{r}_a = \sum_{a=1}^{N} m_a \frac{d^2\vec{r}_a}{dt^2} \cdot \left(\sum_{i=1}^{n} \frac{\partial \vec{r}_a}{\partial q_i} \delta q_i \right)$$

$$= \sum_{i=1}^{n} \underbrace{\left(\sum_{a=1}^{N} m_a \frac{d^2\vec{r}_a}{dt^2} \cdot \frac{\partial \vec{r}_a}{\partial q_i} \right)} \cdot \delta q_i$$

$$\Rightarrow \sum_{a=1}^{N} m_a \frac{d^2\vec{r}_a}{dt^2} \cdot \frac{\partial \vec{r}_a}{\partial q_i} = \sum_{a=1}^{N} \left\{ \frac{d}{dt} \left[m_a \overset{\vec{v}_a}{\frac{d\vec{r}_a}{dt}} \cdot \frac{\partial \vec{r}_a}{\partial q_i} \right] - m_a \overset{\vec{v}_a}{\frac{d\vec{r}_a}{dt}} \cdot \frac{d}{dt}\left(\frac{\partial \vec{r}_a}{\partial q_i} \right) \right.$$

Ejercicio de tarea:

$$\circledast \quad \frac{\partial \dot{\vec{r}}_a}{\partial \dot{q}_i} = \frac{\partial \vec{r}_a}{\partial q_i}$$

$$\vec{V}_a = \dot{\vec{r}}_a = \frac{d\vec{r}_a}{dt} = \frac{d}{dt}\vec{r}_a(q_1, q_2, q_3, \dots, q_n, t)$$

$$= \frac{\partial \vec{r}_a}{\partial q_1}\dot{q}_1 + \frac{\partial \vec{r}_a}{\partial q_2}\dot{q}_2 + \dots + \frac{\partial \vec{r}_a}{\partial q_n}\dot{q}_n + \frac{\partial \vec{r}_a}{\partial t} = \sum_{i=1}^{n} \frac{\partial \vec{r}_a}{\partial q_i}\cdot q_i + \frac{\partial \vec{r}_a}{\partial t}$$

¿Y?

$$\frac{\partial \dot{\vec{r}}_a}{\partial \dot{q}_1} = \frac{\partial \vec{r}_a}{\partial q_1}\frac{\partial \dot{q}_1}{\partial \dot{q}_1}$$; Para cualesquiera q tenemos el siguiente resultado:

$$\frac{\partial \vec{r}_a}{\partial \dot{q}_i} = \frac{\partial \vec{r}_a}{\partial q_i}$$

Delta de Kronecker

$$\delta_{ij} = \begin{cases} 1; & i=j \\ 0; & i \neq j \end{cases}$$

Tenemos

$$\vec{V}_a = \dot{\vec{r}}_a = \frac{d\vec{r}_a}{dt} = \sum_{j=1}^{n} \frac{\partial \vec{r}_a}{\partial q_j}\delta q_j + \frac{\partial \vec{r}_a}{\partial t}$$

Tenemos ...

$$\frac{\partial \vec{r}_a}{\partial \dot{q}_i} = \frac{\partial}{\partial \dot{q}_1}\left[\sum_{j=1}^{n} \frac{\partial \vec{r}_a}{\partial q_j}\dot{q}_j + \frac{\partial \vec{r}_a}{\partial t}\right] = \sum_{j=1}^{n} \frac{\partial \vec{r}_a}{\partial q_j}\frac{\partial \dot{q}_j}{\partial \dot{q}_i}$$

$$= \sum_{j=1}^{n} \frac{\partial \vec{r}_a}{\partial q_j}\delta_{ij} = \frac{\partial \vec{r}_a}{\partial q_i}$$

Ahora...

$$(*)\quad \frac{d}{dt}\left(\frac{\partial \vec{r}_a}{\partial q_i}\right)=\left[\frac{\partial}{\partial q_1}\left(\frac{\partial \vec{r}_a}{\partial q_i}\right)\right]\frac{dq_1}{dt}+\left[\frac{\partial}{\partial q_2}\left(\frac{\partial \vec{r}_a}{\partial q_i}\right)\right]\frac{dq_2}{dt}$$

$$+\dots+\left[\frac{\partial}{\partial q_n}\left(\frac{\partial \vec{r}_a}{\partial q_1}\right)\right]\frac{dq_n}{dt}+\frac{\partial}{\partial t}\left(\frac{\partial \vec{r}_a}{\partial q_i}\right)$$

Tenemos

$$\frac{d}{dt}\left(\frac{\partial \vec{r}_a}{\partial q_i}\right)=\frac{\partial^2 \vec{r}_a}{\partial q_1 \partial q_i}\dot{q}_1+\frac{\partial^2 \vec{r}_a}{\partial q_2 \partial q_i}\cdot\dot{q}_2+\dots+$$

$$\frac{\partial^2 \vec{r}_a}{\partial q_n \partial q_i}\dot{q}_n+\frac{\partial^2 \vec{r}_a}{\partial t \partial q_i}$$

Tenemos

$$=\frac{\partial^2 \vec{r}_a}{\partial q_i \partial q_1}\dot{q}_1+\frac{\partial^2 \vec{r}_a}{\partial q_i \partial q_2}\dot{q}_2+\dots+\frac{\partial^2 \vec{r}_a}{\partial q_i \partial q_n}\dot{q}_n+\frac{\partial^2 \vec{r}_a}{\partial q_i \partial t}$$

$$\vdots$$

$$=\frac{\partial}{\partial q_i}\left[\frac{\partial \vec{r}_a}{\partial q_1}\dot{q}_1+\dots+\frac{\partial \vec{r}_a}{\partial q_n}\dot{q}_n+\frac{\partial \vec{r}_a}{\partial t}\right]$$

Finalmente...

$$\left.\frac{\partial \dot{\vec{r}}_a}{\partial q_i}\right\}$$ Regla de la cadena en varias variables.

Tenemos

$$= \sum_{a=1}^{N} \left\{ \frac{d}{dt}\left[m_a \vec{V}_a \cdot \frac{\partial \vec{V}_a}{\partial \dot{q}_i} \right] - m_a \vec{V}_a \cdot \frac{\partial \vec{V}_a}{\partial q_i} \right\}$$

observación:

$$* \quad \frac{\partial}{\partial \dot{q}_i}\left(\vec{V}_a \cdot \vec{V}_a\right) = \frac{\partial \vec{V}_a}{\partial \dot{q}_i} \cdot \vec{V}_a + \vec{V}_a \cdot \frac{\partial \vec{V}_a}{\partial \dot{q}_i} = 2\, \vec{V}_a \cdot \frac{\partial \vec{V}_a}{\partial \dot{q}_i}$$

Tenemos

$$\vec{r}_a = \vec{r}_a(q_i, t) \quad ; \qquad i = 1, 2, \ldots, 3N = n$$

$$\sum_{a=1}^{N} m_a \frac{d^2 \vec{r}_a}{dt^2} \cdot \frac{\partial \vec{r}_a}{\partial q_i} = \sum_{a=1}^{N} \left\{ \frac{d}{dt}\left[\frac{m_a}{2} \frac{\partial}{\partial \dot{q}_i} \vec{V}_a^{\,2} \right] - \frac{m_a}{2} \frac{\partial}{\partial q_i}\left(\vec{V}_a^{\,2}\right) \right\}$$

$$\vec{V}_a \cdot \vec{V}_a = \vec{V}_a^{\,2}$$

$$= \sum_{a=1}^{N} \left\{ \frac{d}{dt}\left[\frac{\partial}{\partial \dot{q}_i} \underbrace{\left(\frac{1}{2} m_a \vec{V}_a^{\,2} \right)}_{T_a} \right. - \frac{\partial}{\partial q_i} \underbrace{\left(\frac{1}{2} m_a \vec{V}_a^{\,2} \right)}_{T_a} \right\}$$

T_a cinética de la

Partícula a

$$= \sum_{a=1}^{N} \left\{ \frac{d}{dt}\frac{\partial T_a}{\partial \dot{q}_i} - \frac{\partial T_a}{\partial q_i} \right\}$$

$$= \frac{d}{dt}\frac{\partial}{\partial q_i}\underbrace{\left(\sum_{a=1}^{N} T_a\right)}_{T} - \frac{\partial}{\partial q_i}\underbrace{\left(\sum_{a=1}^{N} T_a\right)}_{T}$$

$$= \sum_{a=1}^{N} m_a \frac{d^2\vec{r_a}}{dt^2} \cdot \frac{\partial \vec{r_a}}{\partial q_i} = \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right) - \frac{\partial T}{\partial q_i}$$

$$\Rightarrow \boxed{m_a \frac{d^2\vec{r_a}}{dt^2} = \vec{F}_a} \qquad \{a\}_{i=1}^{N}$$

$$\sum_{a=1}^{N} m_a \frac{d^2\vec{r_a}}{dt^2} \cdot \delta\vec{r_a} = \sum_{a=1}^{N} \vec{F}_a \cdot \delta\vec{r_a}$$

$$\Rightarrow \sum_{i=1}^{n=3N} \left[\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right) - \frac{\partial T}{\partial q_i} \right] \delta q_i = \sum_{i=1}^{n=3N} Q_i\, \delta q_i$$

$$\Rightarrow \boxed{\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right) - \frac{\partial T}{\partial q_i} = Q_i} \qquad i = 1, 2, \dots, n = 3N$$

Fuerza Q_i

$$Q_i \equiv \sum_{a=1}^{N} \vec{F}_a \cdot \frac{\partial \vec{r}_a}{\partial q_i} = \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right) - \frac{\partial T}{\partial q_i} = Q_i \overset{\text{si}}{=} \frac{d}{dt}\left(\frac{\partial U}{\partial \dot{q}_i}\right) - \frac{\partial U}{\partial q_i}$$

$$\Rightarrow \frac{d}{dt}\frac{\partial}{\partial \dot{q}_i}\underbrace{(T-U)}_{\mathcal{L}} - \frac{\partial}{\partial q_i}\underbrace{(T-U)}_{\mathcal{L}} = 0$$

$$\boxed{\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0}$$

<u>Principio de D'alembert</u>

6 – 9 → próximo martes.

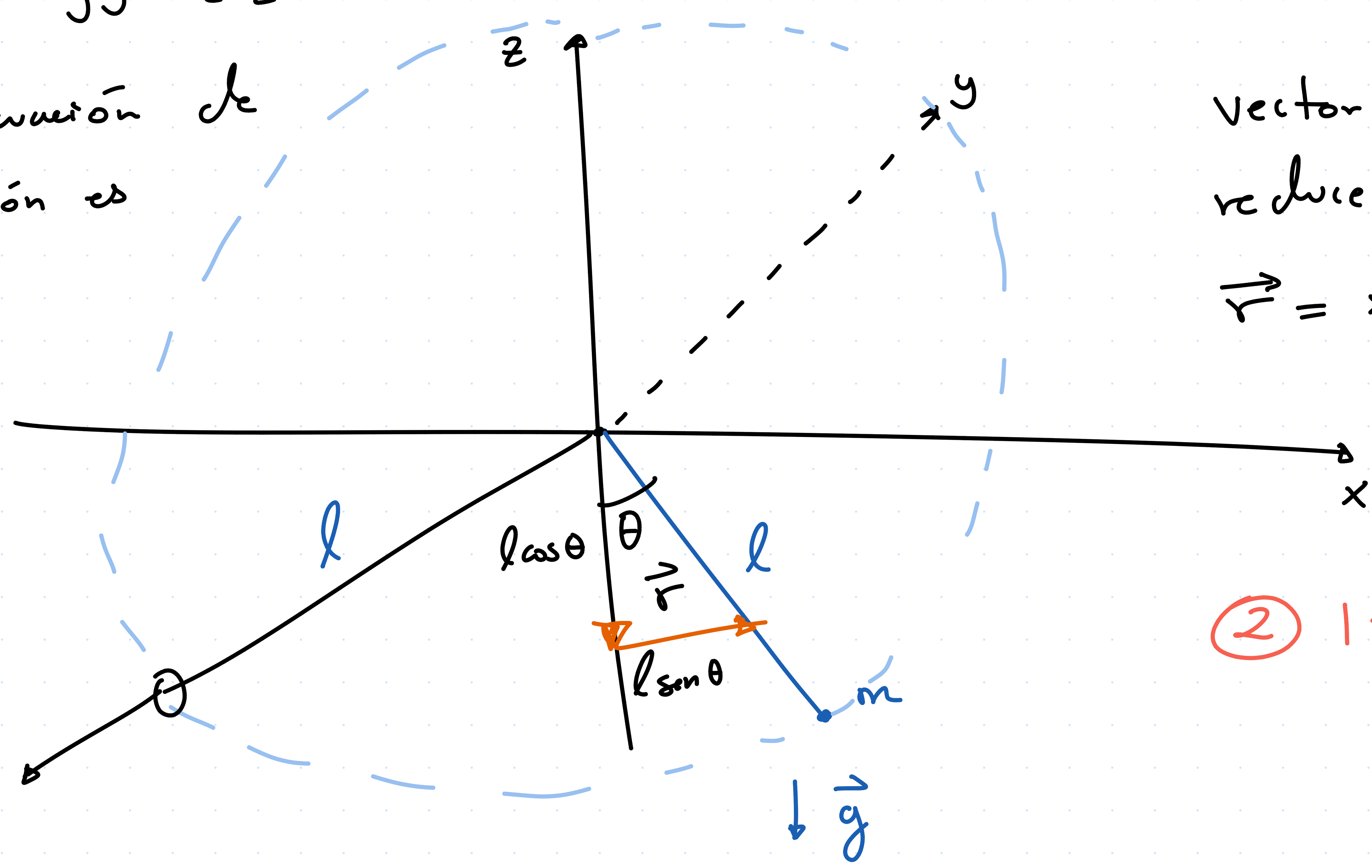
<u>Ejercicio</u>

$\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$

① $y = 0$

- una ecuación de constricción es

$\underline{y = 0}$



vector de posición se reduce a:

$\vec{r} = x\hat{x} + z\hat{z}$

② $|\vec{r}| = \ell$

- Escribir explícitamente las constricciones en la tarea.

$|\vec{r}^{\,2}| = \ell^2$

$|\vec{r}| = \vec{r} \cdot \vec{r}$

$\vdots$

$= x^2 + z^2$

$x^2 + z^2 = \ell^2$

circunferencia de radio ℓ.

- coordenada generalizada sea la más sencilla posible

Posición en coordenadas polares:

$$\boxed{\vec{r} = \ell \operatorname{sen}\theta\,\hat{x} - \ell\cos\theta\,\hat{z}}$$

cuando $\theta = 0$

$\hookrightarrow$ $\underline{\vec{r} = -\ell\,\hat{z}}$

$\vec{r} \cdot \vec{r} = \ell^2(\operatorname{sen}\theta\,\hat{x} - \cos\theta\,\hat{z}) \cdot (\operatorname{sen}\theta\,\hat{x} - \cos\theta\,\hat{z})$

$\vdots$

$= \ell^2(\operatorname{sen}^2\theta + \cos^2\theta) = \ell^2$

velocidad

$$\dot{\vec{r}} = \frac{\partial \vec{r}}{\partial \theta}\dot{\theta} + \cancelto{\vec{0}}{\frac{\partial \vec{r}}{\partial t}}$$

$$\dot{\vec{r}} = \left(l\cos\theta\,\hat{x} + l\,\mathrm{sen}\,\theta\,\hat{z} \right)\dot{\theta}$$

$$\dot{\vec{r}}^{\,2} = l^2\dot{\theta}^2 \quad ; \quad T = \frac{1}{2} m l^2 \dot{\theta}^2 \quad ; \quad L = T - U$$